Content, Source, and Position: The Impacts of Platform-specific Internal Generative AI on Online Q&A Communities

Abstract

The advent of generative artificial intelligence (GenAI) has disrupted online question-and-answer (Q&A) communities and prompted debates on managing AI and human contributions. Unlike prior studies on external general-purpose GenAI systems like ChatGPT, our work analyzes if platform-specific, internally developed GenAI could offer a balanced solution. Leveraging a randomized experiment from a leading Chinese technical Q&A community, we examine human user (answerers and viewers) behaviors after the introduction of internal GenAI. Our findings reveal a reduction in human-provided answers and viewer recognition of these answers. We identify three key mechanisms that drive this reduction: a "content effect," as high-quality AI answers may make human efforts seem redundant; a "source effect," which stems from AI as the information source; and a "position effect," because AI-generated answers appear almost instantaneously in the first position and demotivate human contribution. We show that the reduction in human answer recognition is due to increased expectations for human answers rather than reduced answer quality or expert participation. We further demonstrate that in this AI-augmented environment, internal GenAI prompts human answerers to offer more personalized, experience-based insights. We contribute to the emerging literature on GenAI's impact and provide practical insight for platforms integrating GenAI into online Q&A communities.

Keywords: Generative AI, Human-AI Interactions, Online Communities, Content Effect, Source Effect, Position Effect

1. INTRODUCTION

Generative artificial intelligence (GenAI) and large language models (LLMs) have transformed online information-seeking behavior and present a significant challenge to traditional question-and-answer (Q&A) communities. Recent studies document a decline in user participation on these platforms after the introduction of ChatGPT (Burtch et al. 2023; Quinn & Gutt 2023; del Rio-Chanona et al. 2023; Sanatizadeh et al. 2023; Xue et al. 2023), which raises concerns about the future viability of the online Q&A communities.

In response, many Q&A platforms, such as Quora,[1] now develop their own LLMs that leverage their historical data repositories to answer questions. This integration of customized GenAI within communities (internal GenAI, Figure 1) aims to enhance user experience through timely, high-quality answers that can compete with external GenAI platforms.

A critical distinction exists between external general-purpose GenAI and platform-specific internal GenAI. This distinction has important implications for research methodology, platform dynamics, and stakeholder behaviors that current literature overlooks. External GenAI tools function independently of specific platforms as domain-agnostic all-purpose assistants. These tools divert questions away from online Q&A platforms (Figure 1, left panel), which creates methodological challenges for researchers. The demand diversion remains unobservable within platform data, which precludes a comprehensive assessment of AI's impact on community dynamics. Furthermore, human answerers increasingly use external GenAI when they craft responses. This practice blurs the boundaries between human and AI contributions through hybrid responses that complicate effect attribution and impact assessment.

[1] Further information about the Quora's AI integration can be found at: https://quorablog.quora.com/Poe-1?ch=10&oid=98806369&share=e6baeb55&srid=C5jip&target_type=post.

In contrast, platform-specific internal GenAI operates within the platform's ecosystem under direct governance. This approach retains question demand within the observable platform environment—users post questions on the platform and receive both AI and human responses (Figure 1, right panel). Internal GenAI creates clear demarcations between AI-generated and human-generated content through explicit labeling or distinctive visual styling. Researchers can directly observe which questions receive AI answers, how these compare with human contributions, how humans respond to AI-answered questions, and how viewers engage with different content sources. This transparency provides methodological advantages for the establishment of causal relationships through experimental approaches.

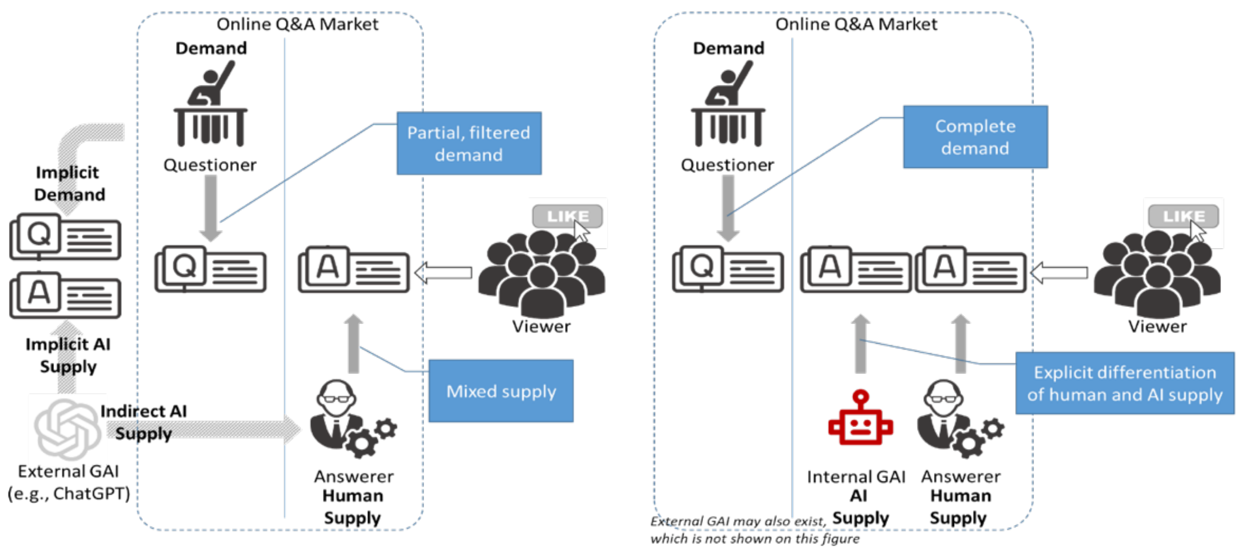

Figure 1. Illustration of online Q&A communities under the impact of GenAI

Our research focuses specifically on internal GenAI systems because they provide relevant insights for platform designers who consider AI integration strategies. The integration of internal GenAI into technical Q&A platforms raises fundamental questions: Will human contributors become discouraged when AI provides instant (high-quality) responses? Will community recognition mechanisms remain effective when both human and AI-generated content coexist? Will human answerers adapt their strategies to differentiate from AI?

While these questions have important implications for the design and sustainability of online Q&A communities, the answers are characterized by theoretical tensions. Although questioners may benefit from faster responses, the effects on answerers and viewers involve competing theoretical predictions. On the one hand, internal GenAI may discourage answerer contributions, as answerers may perceive their contributions as less valuable. We posit three mechanisms that may discourage human contribution: a "content effect" where AI answers make human efforts seem redundant; a "source effect" where AI's perceived authority reduces motivation to contribute; and a "position effect" where AI's occupation of the first-answer position deters subsequent human contributions. On the other hand, alternative perspectives suggest potential benefits: AI answers could relieve the burden on experts by allowing them to focus on more challenging questions where they can add distinctive value. This shift may enhance community-wide specialization and knowledge differentiation. These competing theoretical perspectives highlight the need for empirical investigation to inform both theoretical understanding and practical implementation in this evolving socio-technical environment.

To investigate internal GenAI' impacts, we leverage a randomized experiment conducted by a leading technical Q&A community in China, where the platform introduced a GenAI bot to provide answers to randomly selected questions. This setting is particularly well suited for our study due to the platform's early adoption of internal GenAI, its IT professional user base, and the randomized design that enables clean causal inference.

Our findings reveal a complex landscape of internal GenAI's impacts. We show reduced human answer contributions and provide evidence for the three underlying mechanisms: the content effect, source effect, and position effect. The source effect proves uniquely attributable to AI and does not manifest for human answerers, regardless of their platform reputation, suggesting users potentially overestimate AI capabilities. Internal GenAI also stimulates positive adaptations: human answerers increasingly offer personalized, experience-based insights that emphasize differentiation and highlight distinctive human value. Among content viewers, we observe decreased recognition of human-provided answers. This reduction does not result from lower answer quality or reduced expert participation but rather from increased expectations for human answers in the presence of AI-generated content.

Our paper offers both theoretical and practical contributions. We extend research on external GenAI by investigating platform-specific internal GenAI and its impacts on all key roles in online Q&A communities. We advance understanding of underlying mechanisms, which enables a deeper interpretation of these effects. Our findings provide practical insights to help platforms navigate the challenges and opportunities presented by the AI revolution, informing strategies for integrating AI while maintaining the unique value of human contributions.

2. LITERATURE REVIEW

Our study joins the literature on user engagement and contribution in online Q&A communities. This review synthesizes findings relevant to our study, with emphasis on GenAI impacts on these communities, and identifies knowledge gaps.

2.1. User Engagement and Contribution in Online Q&A Communities

Factors affecting user engagement and contribution in online Q&A communities include question characteristics, feedback and reward mechanisms, and user dynamics and competition.

Question formulation significantly impacts response likelihood and quality. Research has identified several critical factors, including question specificity, linguistic features, and technical presentation. Burke et al. (2010) find that asking specific questions rather than making statements or posing open-ended questions increases response likelihood by 50%. Li et al. (2023)

demonstrate how emotional content and pronoun usage affect answer volume. In technical Q&A communities, Calefato et al. (2018) highlight the importance of concise questions with relevant code snippets and proper formatting for eliciting comprehensive responses. These findings highlight how question structure and content facilitate productive community exchanges.

Existing literature also examines how feedback and rewards motivate contributions. Research on monetary incentives yields mixed results. While some studies suggest positive effects (Berger et al., 2016), others indicate potential negative consequences (Jiang et al., 2024; Wang et al., 2022). This divergence suggests that the effects of monetary reward might be context-dependent. For non-monetary feedback, Moon & Sproull (2008) find that positive feedback motivates participation and reduces user churn, especially among questioners. Li et al. (2020) demonstrate a positive correlation between received answers and subsequent contributions, which emphasizes reciprocity's role in sustaining engagement.

User interactions and relative community standing also impact engagement. Shi et al. (2024) show that informative answers from knowledgeable contributors motivate high-quality contributions from others. Brzozowski et al. (2009) demonstrate that feedback volume correlates strongly with subsequent participation. Based on social comparison theories, Meng et al. (2013) and Chen et al. (2022) show that gamification strategies like hierarchical privilege levels promote knowledge contribution. Wei et al. (2015) reveal that while absolute reputation (e.g., badge count) may not directly motivate contributions, relative reputation (e.g., rankings) generates significant motivation. This finding highlights the importance of social comparison in driving user engagement.

2.2. The Impact of GenAI on Online Q&A Communities

The impact of GenAI on online Q&A communities represents an emerging research stream within technology-mediated user interactions. While existing literature has primarily focused on human participants, GenAI introduces a new paradigm that requires careful examination. Recent work explores GenAI effects across various work domains (e.g., Brynjolfsson et al., 2025; Doshi & Hauser, 2024; Noy & Zhang, 2023), including crowdsourced design platforms (Zhao et al., 2024), digital art platforms (Huang et al., 2023), and AI-assisted code development platforms (Yeverechyahu et al., 2024).

A growing body of literature investigates how general-purpose GenAI, particularly ChatGPT, affects user behavior in online Q&A communities. These studies, summarized in Table 1, focus on questioning and answering activities across three key user groups: questioners, answerers, and viewers. For questioners, consensus exists that ChatGPT has decreased the overall question volume in Q&A communities (Burtch et al., 2023; del Rio-Chanona et al., 2023; Quinn & Gutt, 2023; Sanatizadeh et al., 2023; Xue et al., 2023). This reduction stems primarily from simpler questions now answered by ChatGPT. Consequently, the remaining questions tend to be longer and more complex, which challenges both human and AI respondents (Quinn & Gutt, 2023; Sanatizadeh et al., 2023).

Findings about the impact on answerers are mixed, likely due to the multifaceted nature of GenAI's influence and methodological variations across studies. Some research indicates that ChatGPT can be a valuable tool for answerers by facilitating answer compilation (Li & Kim, 2024; Wang & Zheng, 2024). The reduction in question volume intensifies competition among answerers, potentially leading to a crowding-out effect and prompting remaining participants to enhance the quality of their contributions. This complex interplay has resulted in seemingly contradictory findings regarding answer quantity and quality across different studies. For example, Sanatizadeh et al. (2023) and Li & Kim (2024) find a decrease in the number of answers per question and in the quantity of qualified, undeleted user answers, respectively. In contrast, Wang & Zheng (2024) find that banning ChatGPT-generated content led to a decrease in both answer quantity and the likelihood of answers being accepted by the questioner, suggesting that ChatGPT had a positive effect on these metrics. Shan & Qiu (2023) corroborate this increase in answer volume. The impact of GenAI on answer style is equally diverse. Some studies indicate that external GenAI leads to longer and more complex answers (Borwankar et al., 2023; Sanatizadeh et al., 2023), while others suggest that it facilitates more concise and readable responses (Borwankar et al., 2023; Shan & Qiu, 2023). These conflicting findings may reflect the dual role of GenAI as both a compilation tool and a competition-inducing force driving answerers to differentiate their contributions.

For viewers, research has examined changes in engagement metrics such as views and likes. Sanatizadeh et al. (2023) report a decrease in the number of views following the introduction of ChatGPT, potentially due to users finding answers directly from GenAI without engaging with the online Q&A communities. The impact on likes is less clear, with some studies reporting negative effects (Borwankar et al., 2023; Li & Kim, 2024) and others noting positive outcomes (Sanatizadeh et al., 2023).

Table 1. The impact of generative AI on online Q&A communities

Study	Community	AI Examined	Event	Research Design	User Group Examined	Unit of Analysis	GenAI Effect
Borwankar et al., 2023	Stack Overflow	ChatGPT	GenAI ban	Natural Experiment	Answerer, Viewer	answer-user-day	Make answers shorter, more negative and harder-to-read Decrease # likes
Burtch et al., 2023	Stack Overflow	ChatGPT	GenAI introduction	Natural Experiment	Questioner	topic-week	Reduce # questions
del Rio-Chanona et al., 2023	Stack Overflow	ChatGPT	GenAI introduction	Natural Experiment	Questioner	topic-week	Reduce # questions
Quinn & Gutt, 2023	Stack Exchange	ChatGPT	GenAI introduction	Natural Experiment	Questioner	topic-week	Reduce # questions
Sanatizadeh et al., 2023	Stack Exchange	ChatGPT	GenAI introduction	Natural Experiment	Questioner, Answerer, Viewer	post-day	Reduce # questions; make questions more difficult to read Reduce # answers; make answers longer, more difficult to read Reduce # views; Increase # likes
Shan & Qiu, 2023	Stack Overflow	ChatGPT	GenAI introduction	Natural Experiment	Answerer	user-day	Increase # answers; make answers shorter and easier to read
Su et al., 2023	Quora	Internal GenAI	GenAI introduction	Natural Experiment + Lab Experiment	Answerer	question-week	Increase # answers; make answers longer and similar to AI answers
Xue et al., 2023	Stack Overflow	ChatGPT	GenAI introduction	Natural Experiment	Questioner	topic-day	Reduce # questions; make questions longer, more difficult to read
Li & Kim, 2024	Stack Exchange	ChatGPT	GenAI introduction	Natural Experiment	Answerer, Viewer	user-day	Reduce # answers Reduce # likes
Wang & Zheng, 2024	Stack Exchange	ChatGPT	GenAI ban	Natural Experiment	Questioner, Answerer, Viewer	topic-week	Increase # questions Increase # answers; Increase acceptance ratio of answers Increase # views

2.3. Research Gap

Our literature review (especially Table 1) reveals that existing studies primarily examine the impact of external general-purpose GenAI. There is a notable lack of studies examining the impact of internal GenAI tools provided by online Q&A communities themselves. To date, only Su et al. (2023) directly address this specific issue. Their findings suggest potential positive effects of internal GenAI on user engagement, including increases in the number and length of human-generated answers and greater alignment between human and AI responses. While these insights are valuable, they represent only a first step in understanding the complex interplay between internal AI and user behavior in Q&A communities.

This gap is significant because internal and external GenAI create fundamentally different competitive dynamics. External GenAI competes with platforms for questioners' attention and diverts questions away from the platforms. When external GenAI influences platform content, it does so indirectly: human answerers may incorporate AI-generated content into their responses but obscure its source, which makes AI contributions unobservable and unmeasurable. In contrast, internal GenAI creates direct, observable competition between AI and human answerers within the platform. This transparency allows us to examine the distinct effects of answer content and answerer identity (human versus AI) on user behavior. As organizations across sectors integrate GenAI as supporting tools, understanding how internal AI competes with human contributors becomes increasingly important. Our research addresses this gap by investigating how internal GenAI affects the supply of human answers and the recognition mechanisms (such as upvotes given to answers) that motivate continued participation—elements essential to the sustainability of online Q&A communities.

3. THEORETICAL DEVELOPMENT

3.1. The Impacts of Internal GenAI on Answerers

Prior studies demonstrate that technological interventions can significantly impact participation patterns in online communities (Kraut et al. 2012). Building on this foundation, we propose that internal GenAI reduces human answerers' contributions through three distinct but interrelated mechanisms: the content effect, source effect, and position effect.

3.1.1. The Content Effect

The content effect refers to how AI-generated answers' quality and comprehensiveness influence human answerers' decisions. This mechanism aligns with the Expectancy-Value Theory (Wigfield & Eccles 2000), which suggests individuals engage in behaviors based on success expectations and subjective value assignment.

GenAI systems produce thorough, well-articulated answers that address multiple question aspects (Brown et al. 2020). When potential human answerers encounter such comprehensive AI content, they conduct a mental cost-benefit analysis that evaluates the required effort against the perceived incremental value their contribution would add. Research shows users are motivated by perceptions of unique value addition (Butler et al. 2008; Wasko & Faraj 2005). If users believe an AI-generated answer already provides substantial information, they may conclude their input offers minimal additional value and decide not to contribute.

3.1.2. The Source Effect

The source effect, in contrast, pertains to the effect resulting from the mere fact that the information is generated by an AI system, independent of the content's quality. This effect draws from the automation bias literature, which Mosier et al. (2015) define as people's tendency to over-rely on automated systems and accept their outputs without critical evaluation.

Users may attribute heightened authority or credibility to AI-generated answers due to perceptions of AI's comprehensive knowledge and processing capabilities (Logg et al. 2019). Sundar's (2008) MAIN model explains how technological sources serve as heuristic cues that trigger cognitive shortcuts in information processing. This "machine heuristic" leads users to perceive machine-generated content as more objective and authoritative than human-generated content. Additionally, AI's perceived efficiency creates psychological pressure. Dzindolet et al. (2003) find that when users perceive automated systems as highly efficient, they experience elevated performance expectations. These expectations may discourage human answerers from investing time in crafting answers they believe cannot match AI's perceived efficiency.

3.1.3. The Position Effect

The position effect addresses how AI-generated answers deter human participation by appearing almost instantaneously and occupying the first position. This connects to the "fastest gun in the west" phenomenon in online Q&A communities (Anderson et al. 2012), where users race to be first responders. Anderson et al. (2012) demonstrate that first answers receive disproportionate attention simply by virtue of position and accumulate more votes over time. First answers also remain visible during the critical high-traffic period when questions receive peak attention (Mamykina et al. 2011). Since GenAI produces nearly instantaneous answers, it blocks human users' opportunity to secure the first answer position, which significantly reduces expected rewards crucial for sustained participation.

These mechanisms unite through dual-process theories of information processing. Both the Elaboration Likelihood Model (Petty et al. 1986) and the Heuristic-Systematic Model (Chaiken 1980) posit that people process information through two distinct cognitive pathways: a central route involving careful content consideration and a peripheral route relying on contextual cues and heuristics. The content effect operates primarily through the central route, where users systematically evaluate the AI answer's substance and quality. The source effect functions through the peripheral route, with AI identity serving as a heuristic cue that influences perceptions without necessarily triggering deep content analysis.

The position effect spans both routes. For many users, answer position serves as a peripheral cue for quick decision-making—they choose not to contribute based on the heuristic that non-first answers receive diminished attention without engaging in deeper analysis of potential contribution value. This represents classic peripheral route processing. Simultaneously, other users may engage in systematic processing by consciously assessing the strategic implications of contributing after GenAI occupies the first answer position. These users calculate the diminished visibility their contribution would receive and make deliberate decisions based on cost-benefit analyses. This dual-route activation aligns with Chaiken and Maheswaran's (1994) concept of "multiple-motive" information processing, where the same cue can trigger both heuristic and systematic processing.

This integrated theoretical approach systematically captures internal GenAI's impact on answers. The combined influence of content quality, AI attribution, and temporal positioning creates a comprehensive deterrent effect on human contribution. Therefore, we hypothesize:

H1: Introducing internal GenAI to answer questions reduces the contribution of human answerers in online Q&A communities.

3.2. The Impacts of Internal GenAI on Viewers

We also theorize how internal GenAI influences recognition received by human answers. This dimension is important because recognition, typically through upvotes or other community feedback, constitutes a fundamental reward structure in online Q&A communities (Mamykina et al. 2011). Recognition validates contributors' efforts, signals answer quality to the community and influences future participation decisions (Jin et al. 2015; Luo et al. 2024). We propose an integrated theoretical framework that combines quality-based and expectation-based mechanisms to conceptualize how AI-generated answers affect the recognition of human contributions.

3.2.1. Quality-based Mechanism

The first mechanism posits that AI-generated answers lead to a decline in the actual quality of subsequent human answers through two pathways: expertise composition shift and reduced effort investment.

Expertise Composition Shift: AI-generated answers may disproportionately discourage expert participation while minimally impacting non-expert contribution. Chen et al. (2025) note that experts strategically allocate their attention to questions where they can provide maximum unique value. When an AI system provides a comprehensive answer, experts may perceive their potential contribution as having diminished marginal utility.

This selective participation aligns with "knowledge uniqueness" described by Ludford et al. (2004). Contributors are more motivated when they perceive their knowledge as unique and valuable to the community. When AI answers cover common knowledge aspects, experts may redirect their efforts to questions explicitly requiring specialized knowledge where their expertise would be more distinctly valuable (Khansa et al. 2015).

This expertise migration would result in a compositional shift among human answerers, with the remaining pool containing a higher proportion of non-experts. Consequently, this shift would manifest as reduced overall quality of human answers and subsequently decreased recognition from viewers.

Reduced Effort Investment: AI-generated answers may also decrease answer quality by affecting contributors' effort allocation decisions. Anderson et al. (2012) observe that contributors strategically invest effort based on perceived opportunity for recognition. When an AI system offers an apparently comprehensive answer, human contributors may become disinclined to invest substantial effort in crafting high-quality answers. When contributors perceive that an AI answer has already captured the "low-hanging fruit," they may conclude that significant additional effort would yield only marginal improvements. The perceived effort-to-value ratio may significantly influence contribution quality.

Both pathways—expertise composition shift and reduced effort investment—would result in lower-quality human answers and, consequently, fewer upvotes or other forms of recognition. This quality-based pathway operates through actual changes in the substance of human contributions rather than through changes in evaluation standards.

3.2.2. Expectation-based Mechanism

The second mechanism proposes that human answer quality remains unchanged. Instead, AI-generated answers raise users' expectations for what constitutes a high-quality answer, which leads to more critical evaluations of human contributions.

This mechanism draws upon expectation disconfirmation theory (Oliver 1980), which posits that satisfaction is determined by the gap between expected and perceived performance. When an AI system provides an answer, users may calibrate their expectations to this new standard and anticipate higher levels of quality, accuracy, and comprehensiveness than they would for typical human answers.

Research in human-AI interaction demonstrates that AI systems can significantly influence human expectations. Jussupow et al. (2020) document that when users interact with high-performance algorithm-generated content, they develop heightened expectations for consistency, comprehensiveness, and accuracy.

This expectation-based mechanism operates through changes in evaluation standards rather than through changes in the actual quality of human contributions. The presence of AI-generated answers creates a new reference point against which human answers are judged, potentially leading to more stringent evaluations and reduced recognition.

These two mechanisms—actual quality reduction and expectation elevation—are not mutually exclusive but represent complementary processes that may operate simultaneously. Together, these pathways create a comprehensive theoretical framework that accounts for both supply-side changes (alterations in human contribution quality) and demand-side changes (shifts in evaluation standards). Based on this theoretical foundation, we hypothesize:

H2: Introducing internal GenAI to answer questions reduces the recognition by viewers of human answers in online Q&A communities.

4. RESEARCH CONTEXT

4.1. SegmentFault and a Randomized Experiment on Internal GenAI

Our study examines data from SegmentFault, a prominent Chinese online technical Q&A community that specializes in information technology and software development. Founded in 2012, SegmentFault serves nearly 7 million users and facilitates knowledge exchange across technical domains such as web and mobile app development, artificial intelligence, cloud computing, and cybersecurity. As of April 2024, the platform contains over 310,000 technical questions with an average of 1.9 answers per question. Among global technical Q&A communities, SegmentFault ranks sixth in questions asked, third in answers provided, and first in answer-to-question ratio.[2] It constitutes an essential resource for Chinese developers, programmers, and IT engineers who seek knowledge, solutions, and collaboration opportunities.

SegmentFault's ecosystem relies on user-generated content and interactions. Members pose questions, provide answers, participate in discussions, and express approval through "accept" or "like" actions. To maintain content quality, SegmentFault's operations team reviews user-contributed solutions before publishing them on question pages.

Several features enhance user engagement and content quality. Each answer includes a comment section for additional discussion and idea refinement. A voting system enables questioners to "accept" answers while viewers can upvote or downvote content. SegmentFault also implements a reputation system that rewards members with points and medals based on the quantity and quality of their contributions.

Our study leverages an ongoing randomized experiment conducted by SegmentFault.[3] Beginning September 13, 2023, the platform randomly selected a subset of questions to receive answers from a GenAI bot. According to official platform communication, this randomized experiment aims to gather user feedback and assess whether AI integration enhances overall performance and benefits community members. This experiment offers a unique opportunity to investigate how internal GenAI affects user behaviors and community dynamics. Appendix A provides detailed background information about the experiment.

Figure 2 illustrates the differences between "treated questions" (those with AI-generated answers) and "control questions" (those without). While the platform records precise timestamps for human answers, it omits this information for AI-generated answers. We observe that AI-

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[2] This information can be found at: https://stackexchange.com/sites?view=list#technology-questions.

[3] The randomized experiment announcement can be accessed at: https://meta.segmentfault.com/questions/10010000053637863.

generated answers appear within five minutes after question posting. This response time substantially exceeds the first human answers, which arrive after an average nine-hour delay. Consequently, AI-generated answers typically occupy the first position for questions in the treatment group. AI-generated answers also differ from human answers because they cannot receive endorsements from questioners or votes from viewers.

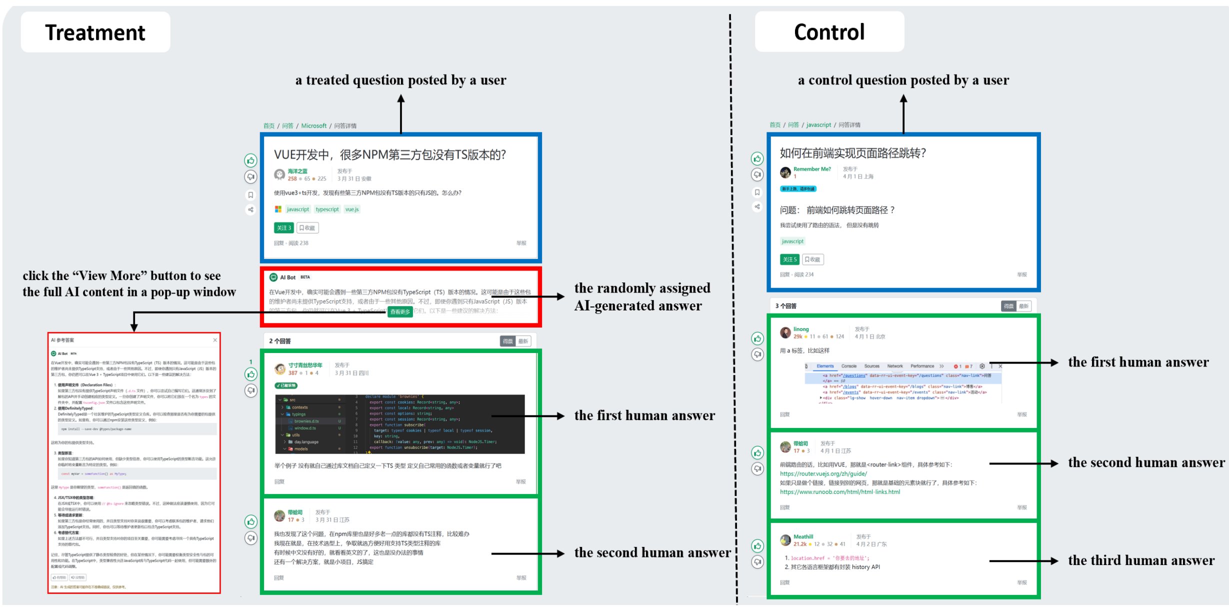

Figure 2. Illustration of treatment and control questions

Community members can provide answers regardless of the presence of an AI-generated answer. When an AI-generated answer exists, viewers typically see this content before human-generated answers. The platform initially displays only two lines of AI-generated content on the question page. Viewers can access the complete answer via a “View More” button, which triggers a pop-up window labeled “AI Answers” with a disclaimer that alerts users to potential errors or inaccuracies and indicates the content is for reference only.

4.2. Data Collection

We collected all questions and answers posted on SegmentFault during a 150-day period from September 2023 (the start of the randomized experiment) to February 2024. Our dataset comprises 3,613 questions posted by 1,404 unique users, averaging 24 new questions per day. AI-generated answers appeared in 1,766 questions (48.9% of the total).

The platform assigns a tag to each question as a content summary. Based on 384 unique tags, we classify questions into ten categories: web and mobile development, software design and architecture, cloud computing, databases, computer networks and systems, AI and machine learning, development tools and environment, programming languages, career and professional development, and others. "Web and mobile development" constitutes the largest category, accounting for 57.5% of the questions in our sample.

4.3. Summary Statistics

The reputation scores in our sample range from -3 to 28,800, with an average of 378.337. This average roughly corresponds to 10 questions, each receiving about 2 upvotes.

Table 2 reports summary statistics of our data. Questions in our sample, on average, have 86.809 Chinese characters. They receive an average of 1.293 human answers, with 76.1% of questions receiving at least one answer. Questioners "accept" answers for 32.2% of all questions.

The treatment group exhibits notable differences compared to the control group. It receives fewer human answers (1.209 vs. 1.372) and has a lower proportion of "accepted" answers (29.3% vs. 35.0%). On question pages, viewers can upvote or downvote answers. The page displays net likes (upvotes minus downvotes) for each answer. Answers in our sample average 0.489 net likes, with the treatment group receiving fewer (0.436 vs. 0.536).

SegmentFault employs a reputation system where users accumulate reputation scores based on their contributions. The reputation scores in our sample range from -3 to 28,800, with an average of 378.337.[4] This average roughly corresponds to 10 questions, each receiving about 2 upvotes.

Table 2. Descriptive statistics of main variables

Variable	Mean and SD			Min	Max
	All samples	Treatment group	Control group		
question length (number of Chinese characters)	86.809 (113.981)	84.233 (89.347)	89.272 (133.323)	0.000	2702.000
number of human-generated answers received by a question	1.293 (1.166)	1.209 (1.133)	1.372 (1.192)	0.000	8.000
percentage of questions that received at least one answer	0.761 (0.427)	0.724 (0.447)	0.795 (0.404)	0.000	1.000
percentage of questions that have an answer "accepted" by the questioner	0.322 (0.467)	0.293 (0.455)	0.350 (0.477)	0.000	1.000
average number of net likes received by all the answers to a question	0.489 (1.096)	0.436 (1.041)	0.536 (1.140)	-3.000	14.000
reputation scores of the questioner at the time the question was posted	378.337 (1135.056)	334.484 (695.343)	420.267 (1433.522)	-3.000	28800.000

Note: Columns 2-4 present the means of the main variables, with the standard deviations shown in parentheses below each mean

4.4. Verifying Random Assignment

Our empirical strategy critically depends on whether the questions are truly assigned to the treatment and control groups at random. To validate this assignment, we analyze the correlation between treatment status and various *pre-assignment characteristics* of both questioners and questions. A truly random assignment should yield no significant correlations between these characteristics and the treatment status.

We use a regression-based approach to assess the random assignment, regressing treatment status on a range of pre-assignment variables, including questioner characteristics, question content features, question types, and timing. Questioner characteristics include reputation scores and medal counts (gold, silver, and bronze), which are components of SegmentFault's user

[4] In SegmentFault, receiving downvotes on an answer can result in deductions to the answerer's reputation score, potentially pushing it into negative territory.

reputation system. Question content features include length (measured by Chinese character count) and the presence of multimedia elements such as photos, hyperlinks, tables, and block quotes. Besides, the questions are in 10 categories, which we consider as question type. To capture question timing, we include a series of binary variables indicating the date on which the question was posted.

Table A in the appendix presents the results of our random assignment test. The first column displays bivariate regressions of treatment status on each pre-assignment characteristic individually. Columns 2-4 progressively incorporate questioner characteristics, question content features, and question type into multivariate regressions, all including question timing fixed effects. We find that none of the pre-assignment characteristics are statistically significant predictors of treatment status, and the pre-assignment characteristics explain very little of the variation in the treatment status, as evidenced by the extremely small adjusted R-squared value. These results provide evidence for the validity of random assignment.

5. EMPIRICAL STUDY

5.1. Main Variables of Interest

Our study investigates the impact of internal GenAI on online Q&A communities. We focus on *answer supply* and *view recognition*. To assess the impact on answer supply, we analyze the number of human-provided answers per question. Our primary measure focuses on answers received within the first 7 days after posting, as approximately 80% of questions in our sample received all their answers within this timeframe. This approach mitigates potential outliers from long-tail effects where some questions continue receiving answers long after posting. For robustness, we also consider the total number of human answers as an alternative measure.

We measure viewer recognition through net likes. Users can upvote or downvote answers on SegmentFault. The difference between these votes (net likes) serves as a proxy for answer quality and community appreciation. This measure allows us to assess how AI-generated answers affect the recognition of human contributions.

5.2. Econometric Models

To examine the impact of AI-generated answers on the supply of human answers, we run the following regression model:

$$answerNum_i = \beta_0 + \beta_1 AI_i + \gamma X_i + questionTiming_i + \varepsilon_i \tag{1}$$

Here, i indexes questions. The dependent variable, $answerNum_i$, is the logarithm of the number of human answers received by question i. AI_i is a binary variable indicating whether the question was treated, that is, provided with an AI-generated answer ($AI_i = 1$ if treated, otherwise $AI_i = 0$). X_i is a vector of control variables, including the pre-assignment characteristics of the questioner, question content, and question type, as described in the earlier section on verifying random assignment. We also include question timing fixed effects ($questionTiming_i$), indicating the date on which question i was posted. The coefficient of interest, β_1, captures the effect of AI-generated answers on the number of human answers. In this analysis, standard errors are clustered at the question tag level to allow for correlations between questions with the same tag.[5]

To examine the impact of AI-generated answers on the recognition of human answers, we run the following regression model:

$$netLike_{ij} = \beta_0 + \beta_1 AI_i + \sum_{s=1}^{2} \tau_s\, age_{ij}^{s} + \gamma X_i + \varepsilon_{ij} \tag{2}$$

[5] Clustering the standard errors at other levels, such as date, day of week, and month of year when the question was posted, does not qualitatively change our results.

The dependent variable, $netLike_{ij}$, is net likes for the j^{th} answer to question i. As answers submitted earlier have more time to accumulate likes, we control for answer age (age_{ij}) in days since submission. To account for the potential non-linear accumulation of likes over time, we include a quadratic polynomial of answer age. X_i is a vector of control variables specified in Equation 1. The coefficient of interest, β_1, captures how AI-generated answers affect viewers' recognition of human answers. Standard errors are clustered at the question level to allow for correlations among answers to the same question.

6. RESULTS

6.1. Main Results

Table 3 shows the impact of AI-generated answers on human answer supply. Column 1 reports answers within the first 7 days after question posting. Columns 2-4 show answers within 1 day, 14 days, and all available answers in our sample, respectively.

The presence of AI-generated answers significantly reduces human-generated answers. Column 1 indicates a statistically significant reduction (β = -0.046, p < 0.001), representing a 4.5% decrease ($e^{-0.046}-1$) in human answers. Columns 2-4 show consistent effects with estimates of -0.041, -0.049, and -0.068, all statistically significant. This consistency across different timeframes confirms the robustness of our results.

Table 3. The impact of AI-generated answers on the number of answers provided by humans

DV: $answerNum_i$	(1) Within 7 days	(2) Within 1 Day	(3) Within 14 days	(4) All answers
AI_i	-0.046***	-0.041***	-0.049***	-0.068***
	(0.011)	(0.011)	(0.011)	(0.017)
question timing FE	√	√	√	√
other controls	√	√	√	√
N	3,613	3,613	3,613	3,613

Notes: 1. ***p<0.001; **p<0.01; *p<0.05.
2. Robust standard errors are clustered at the question tag level.
3. Question timing FE indicates the date on which the question was posted.

Table 4 shows the impact of AI-generated answers on answer recognition measured by net likes received by human answers. Column 1 presents baseline results without answer age controls. Column 2 adds a cubic polynomial of answer age to account for non-linear trends in like accumulation over time. Column 3 uses answer time fixed effects to compare treated and control answers posted on the same day.

All specifications show that answers in the treatment group (questions with AI-generated answers) receive significantly fewer likes than the control group. The control group averages 0.311 net likes per answer. Column 2 estimates a decrease of 0.046 net likes due to AI-generated answers. This represents a substantial 14.8% (0.046/0.311*100%) reduction in net likes.

Table 4. The impact of AI-generated answers on recognition by viewers

DV: $netLike_{ij}$	(1)	(2)	(3)
AI_i	-0.045* (0.023)	-0.046* (0.023)	-0.048* (0.022)
controlling for a quadratic polynomial of answer age		√	
answer timing FE			√
other controls	√	√	√
N	4,670	4,670	4,670

Notes: 1. ***p<0.001; **p<0.01; *p<0.05.
2. Robust standard errors are clustered at the question level.
3. Answer timing FE indicates the date on which the answer was posted.

6.2. Robustness Checks

6.2.1. Robustness to Potential Spillover Effects

On SegmentFault, AI-generated answers are visible to all users and could potentially divert attention from questions without AI answers. This contamination might influence outcomes for control questions based on the treatment status of other questions. Such spillover effects would violate the Stable Unit Treatment Value Assumption (SUTVA), which requires that potential outcomes depend exclusively on a unit's own treatment status—a fundamental condition for valid causal inference in randomized experiments.

To test for potential spillover effects, we implement a comparative analysis between our control group and historical data from before the introduction of AI-generated answers (May 2023 - September 2023). This five-month historical period matches the five-month span of our control group. This approach provides a clear test for spillover effects. If spillover effects exist, control questions during the experiment period should show different outcomes compared to similar questions from the pre-AI period, as the control questions would be indirectly affected by AI answers to treated questions. For this analysis, we replace treatment questions with questions posted during the five months before the randomized experiment and rerun Equations 1 and 2. We substitute question timing fixed effects in Equation 1 with a quadratic polynomial of question age controls (the duration from posting until our data collection date) because the two groups have no temporal overlap to allow for question timing fixed effects.

The results in Columns 1 and 2 of Table 5 show no statistically significant differences between our control group and data from the period before AI-generated responses for our two main dependent variables. This absence of significant differences suggests that spillover effects might not be a concern in our experimental setting.

Table 5. Robustness to potential spillover effects

	(1) DV: $answerNum_i$	(2) DV: $netLike_{ij}$
post-AI deployment control group indicator	-0.025 (0.015)	-0.036 (0.027)
controlling for a quadratic polynomial of question age	√	
controlling for a quadratic polynomial of answer age		√
other controls	√	√
N	21.968	11,588

Notes: 1. ***p<0.001; **p<0.01; *p<0.05.
2. Robust standard errors are clustered at the question tag level.

The absence of spillover effects likely results from the platform's interface design. Users first see a list of question titles without any indication about which questions have received AI-generated answers. Users discover AI-generated answers only after they select a specific question title and navigate to the detailed page with question content and answers. This two-step discovery process creates a natural barrier that likely prevents any influence of treatment status on browsing behavior at the question selection stage. Since users cannot distinguish between treatment and control questions during initial browsing, their attention allocation remains unbiased by the presence of AI answers. This interface design preserves the independence of potential outcomes as required by SUTVA.

6.2.2. Robustness to Quality of Random Assignment

Since we did not conduct the experiment, we lacked direct control over the random assignment process. Although our tests show no relationship between treatment status and observable question and questioner characteristics, the selection of unobservables remains a concern—the platform might assign AI answers based on internal metrics unobservable to us.

To address this concern, we employ the Oster method (Oster, 2019), which assesses result robustness when true random assignment might be compromised by the selection of unobservables. This method examines how the estimated treatment effect changes with observable controls and infers potential changes if we could control for unobservables.

The Oster method provides insights through two key parameters. The sensitivity parameter δ indicates how important unobservables would need to be, relative to observables, to explain away the estimated treatment effect. A large δ suggests that unobservables would need to be much more important than observables to nullify the treatment effect. Generally, a δ greater than 2 indicates results robust to selection on unobservables. The second parameter, β^*, is the bias-adjusted effect, which estimates the true treatment effect if selection on unobservables were present. These parameters require an assumption on R^2_{max}, the maximum possible R-squared value achievable if all unobservables were controlled for. R^2_{max} is typically set at 1.3 times the R^2 from the regression with observable controls only (Oster, 2019).

Our analysis in Table 6 reveals two important results. First, we observe remarkable consistency between the original coefficients and the bias-adjusted treatment effects (β^*) across our key dependent variables. For *answerNum*$_i$, the original coefficient of -0.046 differs only marginally from the bias-adjusted coefficient of -0.045. Similarly, for *netLike*$_{ij}$, we observe a negligible difference between the original (-0.048) and bias-adjusted (-0.047) coefficients. Second, the δ values for *answerNum*$_i$ and *netLike*$_{ij}$ are strikingly large (33.868 and 38.867, respectively). These δ values far exceed the recommended threshold of 2 (Oster, 2019). The results provide compelling evidence for the robustness of our results and the quality of the random assignment. This suggests that it is highly unlikely for unobserved confounders to completely negate or substantially change our main conclusions regarding the impact of AI-generated answers on the supply and recognition of human-generated answers.

Table 6. Robustness to the quality of random assignment

	Original coeffient $\hat{\beta}$	R^2	$R^2_{\max}$	Bias-adjusted effect β^*	δ for β=0 given $R^2_{\max}$
DV: *answerNum*$_i$	-0.046	0.053	0.069	-0.045	33.868
DV: *netLike*$_{ij}$	-0.048	0.022	0.029	-0.047	38.867

Note: $\hat{\beta}$ and R^2 refer to our OLS coefficients and corresponding R-squared values. β^* is the bias-adjusted treatment effect. δ indicates how important unobservables would need to be, relative to observables, to explain away the estimated treatment effect.

6.2.3. Alternative Model Specification

One of our dependent variables, the number of answers per question, is count data. We used a log transformation of this variable in our main analysis. Recent research has raised concerns about the practice of log-transforming nonnegative dependent variables (Chen & Roth 2024). Therefore, we ran a Poisson regression using pseudo maximum likelihood (PPML) as an alternative specification. Table 7 Column 1 shows that the result remains consistent with our main results. This analysis confirms that our results are robust across different specifications.

Table 7. Robustness check for alternative model specifications

	(1)
	DV: *answerNum*$_i$ (the count form for Poisson regression)
AI_i	-0.122***
	(0.030)
other controls	√
question timing FE	√
N	3,613

Notes: 1. ***p<0.001; **p<0.01; *p<0.05.
2. Robust standard errors are clustered at the question tag level.
3. Question timing FE indicates the date on which the question was posted.

6.3. Mechanisms

Our results reveal the negative effects of AI-generated answers on the supply and recognition of human-generated answers. In this section, we investigate the mechanisms underlying these effects and discuss their implications.

6.3.1. Testing Mechanisms Driving Reduction in Human-provided Answers

We identified three potential mechanisms behind reduced human answers: content, source, and position effects. To test the content effect, we added content characteristic variables to our main specification. If AI answers deter human participation primarily through their content quality, these content characteristic variables should capture this effect, and the direct effect of the AI indicator should diminish as these content variables explain the influence.

We consider several content characteristics in this analysis. First, we include the length of the answer as a simple proxy for quality and informativeness. Second, we employ a large language model (LLM) to evaluate answer content quality (Li et al. 2024). The detailed method for generating this content quality measure and its validity assessment is provided in Appendix B. Third, a single quality rating may inadequately capture answer quality due to its multidimensional nature. We therefore expand our analysis to include multiple established quality dimensions from the literature: clarity (Rath et al. 2017), readability (John et al. 2011; Podgorny 2016), and accuracy (Blooma et al. 2012; John et al. 2011). Through expert consultations in human-AI interaction research, we further identify seven additional dimensions: relevance, level of detail, personal experiences, personal insights, innovative approaches, alternative solutions, and engaging storytelling. Appendix B provides complete definitions for these dimensions (each on a 1-10 scale).

Table 8 presents our results. Column 1 restricts the sample to questions with at least one answer, which enables control for the first answer's content characteristics. This specification shows a stronger effect of AI-generated answers on human answer quantity (β = -0.157, p < 0.001) compared to our main models. Column 2 adds the logarithm of the first answer length, and Column 3 incorporates an LLM-enabled content quality rating. These controls reduce the AI_i coefficient substantially (from -0.157 to -0.112 and -0.114, respectively). The coefficient for answer length is statistically significant (β = -0.023, p < 0.001), while the LLM-enabled content quality rating is not (p = 0.195). We conducted a supplementary analysis with ten specific LLM-enabled content quality dimension ratings. Column 4 demonstrates that even with these granular quality measures, the coefficient of AI_i remains qualitatively unchanged (β = 0.112).

Table 8. The content and source effects of AI-generated answers

	(1)	(2)	(3)	(4)	(5)	(6)
	answerNum					
AI_i	-0.157***	-0.112***	-0.114***	-0.112***	-0.126***	-0.089***
	(0.012)	(0.016)	(0.018)	(0.017)	(0.008)	(0.017)
$length1Ans_i$		-0.020***	-0.023***	-0.028***		-0.025***
		(0.004)	(0.004)	(0.005)		(0.006)
$AIRaiting1Ans_i$			0.005	-0.001		0.001
			(0.003)	(0.004)		(0.005)
$clarity1Ans_i$				-0.007		-0.006
				(0.009)		(0.008)
$readability1Ans_i$				0.002		0.001

	(1)	(2)	(3)	(4)	(5)	(6)
				(0.008)		(0.007)
$\mathit{accuracy1Ans}_i$				0.009		0.006
				(0.005)		(0.005)
$\mathit{relevance1Ans}_i$				-0.007		-0.007
				(0.006)		(0.006)
$\mathit{detail1Ans}_i$				0.003		0.006
				(0.008)		(0.009)
$\mathit{experience1Ans}_i$				-0.019**		-0.014*
				(0.006)		(0.006)
$\mathit{insight1Ans}_i$				0.016***		0.012**
				(0.005)		(0.004)
$\mathit{innovative1Ans}_i$				0.005		0.005
				(0.007)		(0.006)
$\mathit{alternative1Ans}_i$				0.003		0.000
				(0.006)		(0.006)
$\mathit{storytelling1Ans}_i$				-0.002		-0.001
				(0.007)		(0.006)
$\mathit{firstAnsFromFirstPositionSeeker}_i$					0.178***	0.176***
					(0.016)	(0.015)
only include questions with at least one answer	√	√	√	√	√	√
add the length of the first answer		√	√	√		√
add a single LLM-enabled content quality ratings for the first answer			√	√		√
add ten specifc LLM-enabled content quality dimension ratings for the first answer				√		√
add whether the first human answer was provided by first position seeker					√	√
question timing FE	√	√	√	√	√	√
other controls	√	√	√	√	√	√
N	3,235	3,235	3,216	3,216	3,235	3,216

Notes: 1. ***p<0.001; **p<0.01; *p<0.05.
2. Robust standard errors are clustered at the question tag level.
3. Question timing FE indicates the date on which the question was posted.
4. Columns 3,4 and 6 have smaller samples because our LLM was unable to assess certain answers involving terminologies about network technologies (such as VPN). We exclude those answers in the analysis.

The position effect suggests that an AI answer in the first position of the answer sequence deters participation from users who prefer to provide first answers (hereafter "first position seekers"). Similar to our content effect testing approach, we examine the position effect by observing changes in the AI_i coefficient after accounting for the presence of these position-motivated answerers. First, we identify "first position seekers" using pre-experiment data (May-

September 2023). For each user, we calculate a "first-answer seeking tendency"—the proportion of their answers that appear in the first position relative to their total answer contributions. This metric has a median of 0.417, with the 25th percentile at 0 and the 75th percentile at 1. We designate users with above-median tendencies as "first-position seekers." [6] Then, we include a variable $firstAnsFromFirstPositionSeeker_i$ in our analysis to indicate whether the first human answer to question i came from a "first-position seeker." If the position effect drives the AI treatment effect, incorporating this variable should substantially reduce the AI treatment indicator coefficient, suggesting that AI answers primarily reduce human participation by deterring users who specifically value claiming the first position.

As shown in Table 8 Column 5, adding the $firstAnsFromFirstPositionSeeker_i$ reduces the AI_i coefficient from -0.157 (Column 1) to -0.126, representing approximately a 20% reduction in effect size. The coefficient of AI_i, while reduced, remains statistically significant. These results suggest the position effect partially explains the reduction in human answers. Importantly, the inclusion of content characteristics and $firstAnsFromFirstPositionSeeker_i$ does not render the coefficient of AI_i nonsignificant (Column 6). This pattern of results leads us to conclude that the negative effect of AI-generated answers is due to all three types of effects.

6.3.2. Testing Mechanisms Driving Reduction in Answer Recognition

In this section, we investigate why the presence of AI-generated answers lowers recognition of human answers, as indicated by the fewer net likes received. We suggest two mechanisms that could potentially drive this effect: a quality-based mechanism and an expectation-based mechanism.

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[6] Using the mean instead of the median as the threshold for identifying "first position seekers" yields quantitatively similar results.

The quality-based mechanism posits that AI-generated answers lead to a decline in the actual quality of subsequent human answers. This quality reduction can manifest through two different pathways: expertise composition shift and reduced effort investment.

To test whether the presence of AI-generated answers changes the composition of subsequent human answerers, we construct a measure of answerers' expertise level based on the number of answers contributed by these answerers that are "accepted" by the questioners. The rationale behind choosing the number of "accepted" answers as a measure of expertise, rather than using reputation scores or the total number of net likes received, is that when a questioner accepts an answer, it directly signifies that the answer provided is of the highest quality among the available answers and most helpful in addressing the specific question asked. It is a clear indication that the user has demonstrated expertise in the particular domain or topic related to the question. This validation by the questioner, who is most familiar with the question and its context, carries more weight than metrics such as upvotes or reputation scores, which can be influenced by factors other than expertise.

We test the potential expertise composition shift by investigating the association between the presence of an AI-generated answer and the average expertise level of the human answerers. If the presence of an AI-generated answer changes the composition of answerers by discouraging the participation of expert answerers, we would observe a decline in the expertise scores when an AI-generated answer is provided. We run the following regression:

$$expertise_{ij} = \beta_0 + \beta_1 AI_i + \gamma X_i + answerTiming_{ij} + \varepsilon_{ij} \tag{3}$$

Here, subscript i indexes individual questions, and j indexes the answers to question i. $expertise_{ij}$ is the expertise score of the answerer who contributes the j^{th} answer to question i. X_i is a vector of control variables as specified in Equation 1. $answerTiming_{ij}$ is the answer timing fixed effects, accounting for the date on which the answer is posted. The coefficient of interest is β_1, which indicates whether the answerers in the treatment group have lower expertise scores compared to those in the control group. Standard errors are clustered at the question level. The results are summarized in Table 9 Column 1.

The results indicate that the presence of an AI-generated answer is not associated with lower expertise scores of the human answerers (β = 0.090, p = 0.180). While we proposed that the presence of AI-generated answers might disproportionately discourage expert participation, leading to a shift in the composition of human answerers toward a greater proportion of novices, the results indicate an opposite effect, although the effect is not statistically significant.

We then test whether the presence of AI-generated answers lowers the underlying quality of subsequent human answers. We regress the LLM-enabled content quality ratings of the answers on the binary variable, indicating whether the corresponding question received an AI-generated answer. The rationale behind this test is that if the presence of an AI-generated answer leads to a decline in the number of upvotes for human answers due to a decrease in the underlying quality of these answers, we would observe a corresponding drop in the LLM-enabled content quality ratings when an AI-generated answer is provided. Specifically, we run a modified version of Equation 3, where the dependent variable is replaced with $AIRating_{ij}$, which represents the LLM-enabled content quality rating of the j^{th} answer to question i. The coefficient of interest is β_1, which now indicates whether the answers in the treatment group have lower LLM-enabled content quality ratings.

The results are summarized in Column 2 of Table 9. We find that the presence of an AI-generated answer is not associated with lower LLM-enabled content quality ratings on the human answers (β = -0.002, p = 0.975). As a robustness check, we replace the dependent variable with the logarithm of the length of the answer (*$length_{ij}$*), considering that length of the answer is an indicator of comprehensiveness and informativeness. The results in Column 3 show that the human answers are longer when there is an AI-generated answer, but this effect is not statistically significant (β = 0.076, p = 0.072). Taken together, these results do not support the mechanism that the presence of AI-generated answers lowers the underlying quality of subsequent human answers.

Table 9. Test for answerer compositional change or decline in underlying answer quality

	(1) DV: expertise scores of the answerer who contributes the j^{th} answer to question i ($expertise_{ij}$)	(2) DV: LLM-enabled content quality ratings of the j^{th} answer ($AIRating_{ij}$)	(3) DV: the logarithm of length of the j^{th} answer ($length_{ij}$)
AI_i	0.090	-0.002	0.076
	(0.067)	(0.068)	(0.042)
answer timing FE	√	√	√
other controls	√	√	√
N	4,670	4,631	4,670

Notes: 1. ***p<0.001; **p<0.01; *p<0.05.
2. Robust standard errors are clustered at the question level.
3. Answer timing FE indicates the date on which the answer was posted.

Next, we test the expectation-based mechanism (whether AI-generated answers raise users' expectations of answer quality, leading to fewer upvotes given to the subsequent answers). According to this explanation, users become more critical in their evaluation of human answers because they expect these answers to offer additional insights that complement or expand upon the AI-generated content. If this explanation holds true, we would expect the negative effect on the net likes received by human answers to become more pronounced when the AI-generated answer is of superior quality. This is because a high-quality AI-generated answer would raise user expectations to such an extent that it becomes increasingly difficult for human answers to meet or exceed these elevated expectations by providing additional insights.

To test this mechanism, we first create two subgroups of questions from the treatment group based on the quality of the first (AI-generated) answer. We quantify answer quality using

LLM-enabled content quality ratings. From the 3,613 questions in the treatment group, we identify 993 questions with an LLM-enabled content quality rating in the top 25% quantile (scores ranging from 8 to 10), indicating the superior quality of the first answer. Conversely, another 993 questions display quality ratings in the bottom 25% quantile (ratings ranging from 0 to 6), denoting the low quality of the first answer. We then extract all human answers under these two subgroups of questions for subsequent analysis.

Next, we conduct an answer-level analysis using the model specified in Equation 2, which is applied separately to these two subsamples. Our model controls for a cubic polynomial of answer age and pre-assignment characteristics of the questioner, question content, and question type. The results of this analysis are presented in Table 10, with Column 1 representing the subsample where the first answer is of high quality and Column 2 representing the subsample where the first answer is of low quality.

When the quality of the AI answer is high (Column 1), the presence of an AI-generated answer significantly lowers the number of net likes received by human answers ($\beta = -0.174$, $p < 0.001$). However, when the quality of the AI answer is relatively low (Column 2), this negative effect disappears ($\beta = 0.037$, $p = 0.391$).

These results lend strong support to our prediction that when the AI-generated answer is of high quality, human answers are more likely to be perceived as inadequate or lacking in value-added content, leading to a significant reduction in the net likes received. Conversely, when the quality of AI-generated answers is low, users' expectations are likely to remain grounded, mitigating the negative impact on the evaluation of human answers.

Table 10. Test for the expectation-based mechanism

	(1)	(2)
	DV: $netLike_{ij}$	
AI_i	-0.174***	0.037

	(0.041)	(0.043)
controlling for a cubic polynomial of answer age	√	√
other controls	√	√
N	1,697	1,945

Notes: 1. ***p<0.001; **p<0.01; *p<0.05.
2. Robust standard errors are clustered at the question level.

7. DISCUSSION

7.1. Is There an Illusion of Authority for AI-generated Answers?

We investigate whether the source effect observed for AI-generated answers also exists for high-reputation human contributors. This comparison reveals whether decreased participation stems from general deference to perceived authority (which would apply to high-reputation humans as well) or represents a specific response to AI. Our analysis helps determine if an "illusion of authority" exists uniquely for AI-generated content—a psychological phenomenon where users attribute excessive credibility to AI regardless of content quality.

SegmentFault assigns reputation scores to users based on their contributions. These scores appear beneath usernames and signal credibility and expertise in the community. Due to its visual salience, it is theoretically plausible that the reputation score might generate an effect similar to the source effect of AI-generated answers. To compare the AI source effect with potential answerer reputation effects, we use the following regression model:

$$answerNum_i = \beta_0 + \beta_1 AI_i + \beta_2\, reputation_i + \beta_3\, contentQuality_i + \gamma X_i + questionTiming_i + \varepsilon_i \quad (4)$$

where i indexes individual questions, and AI_i indicates whether question i received an AI-generated answer. $reputation_i$ is the first human answerer's reputation score, measured at the time when their answer was provided. The variables AI_i and $reputation_i$ are complementary indicators that collectively capture the identity and characteristics of the first answerer. These variables cannot simultaneously take positive values. If $AI_i = 1$, indicating an AI-generated first answer, then $reputation_i$ is necessarily 0, reflecting the absence of a human user as the first answerer, and vice versa. By designing these variables in this mutually exclusive manner, we can compare the source effect of AI-generated answers (captured by β_1 when content quality is controlled for) against the reputation effect of human answerers (reflected in β_2). Other variables on the right-hand side of the equation are the same as specified in Equation 1.

Table 11 presents our results. To facilitate comparison, Column 1 reproduces the previously established AI source effect (β = -0.114, $p < 0.001$). To investigate the potential reputation effect of human answerers, we use several specifications. Column 2 introduces $reputation_i$ as a linear term. This analysis reveals no significant effect of the first human answerer's reputation score on subsequent human answer supply. We extend our analysis in Column 3 by incorporating a cubic polynomial of $reputation_i$. However, this more flexible specification also fails to yield a statistically significant reputation effect. In Column 4, we use a binary indicator for reputation exceeding 48,911. This threshold is chosen based on our analysis showing that human answerers with reputation scores ≥ 48,911 generate answers of comparable quality to those produced by the AI system, as measured by LLM-enabled content quality ratings (confirmed by a t-test showing no statistically significant difference). Yet even these highly reputable answerers fail to produce an effect similar to the AI source effect.

Table 11. Comparing the effects of different human reputation levels with the AI source effect

	(1)	(2)	(3)	(4)
	DV: $answerNum_i$			
AI_i	-0.114*** (0.018)	-0.117*** (0.017)	-0.114*** (0.019)	-0.114*** (0.018)
$reputation_i$		-0.000 (0.000)	0.000 (0.000)	
$reputation_i^2$			-0.000 (0.000)	
$reputation_i^3$			0.000 (0.000)	
$I(reputation_i > 48{,}911)$				0.018 (0.068)

controlling for LLM-enabled content quality ratings and length of the first answer	√	√	√	√
question timing FE	√	√	√	√
other controls	√	√	√	√
N	3,216	3,216	3,216	3,216

Notes: 1. ***p<0.001; **p<0.01; *p<0.05.
2. Robust standard errors are clustered at the question tag level.
3. Question timing FE indicates the date on which the question was posted.

Our results consistently demonstrate that while AI-generated answers significantly reduce subsequent human answers, human answerers' reputation shows no significant effect on answer supply. This finding holds regardless of how we model reputation—linearly, cubically, or as a binary variable for highly reputable answerers. All models control for first answer quality to ensure observed effects are not due to content quality variations.

These findings suggest "over-recognition" of AI-generated answers. Users respond differently to AI-generated answers than human-generated answers, even when content quality is held constant. The source effect is unique to AI and is not observed even by highly reputable human answerers capable of producing comparable content quality. The results suggest that an "illusion of authority" may exist uniquely for AI-generated content.

7.2. How Do Answerers Respond to AI-generated Answers?

Our analysis reveals two key challenges facing human answerers in online Q&A communities influenced by internal GenAI: deterred answer supply due to diminished perceived value of human contributions and elevated quality expectations from viewers. This raises the question: How do human answerers respond to these challenges?

We posit that human answerers may adopt differentiation strategies to distinguish their contributions from AI-generated content. This approach is particularly viable in technical domains for two reasons. First, the structured nature of technical questions—typically concise with code snippets—enables answerers to anticipate and customize responses based on established norms (Calefato et al. 2018). Second, AI-generated answers create an opportunity for humans to provide complementary insights that demonstrate their unique value through domain expertise and contextual understanding.

To investigate whether people differentiate their answers in the presence of AI-generated answers, we measure text similarity between first and second answers using Sentence-BERT architecture (Reimers & Gurevych 2019).[7] We then examine the impact of AI-generated answers on this text similarity. As shown in Table 12, when the first answer is AI-generated, the similarity between the first and second answers decreases significantly (β = -0.034, $p < 0.001$), representing a 0.202 standard deviation change. One potential concern is that text length might systematically influence similarity calculations. As the first AI answer in the treatment group tends to be lengthy compared to the first human answer in the control group, we control for LLM-enabled content quality ratings and the first answer length in Column 2. This adjustment reveals an even stronger effect (β = -0.086, $p < 0.001$). This finding confirms that human answerers do differentiate their answers in the presence of AI-generated answers.

To examine specific differentiation dimensions that might benefit viewers, we identify three areas where humans potentially have unique advantages: incorporation of personal experiences, expression of opinionated insights, and suggestion of alternative solutions. To quantify these dimensions, we use an LLM that we instruct to rate answers on a scale of 0-10 for the presence of personal experiences, opinionated insights, and alternative solutions. Given the prevalence of zero scores, we dichotomize these variables, categorizing answers scoring 0 as not

[7] We choose a fine-tuned SentenceBERT model from the official SentenceBERT model collection (https://www.sbert.net/docs/pretrained_models.html). This model, called all-MiniLM-L6-v2, is specifically recommended by the SentenceBERT team for calculating semantic similarity between sentences (https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2).

containing the element and all others as containing it. Using the specification from Equation 2, we regress these binary outcomes on treatment status.

The results, reported in Columns 3 to 5 of Table 12, indicate that AI-generated answers significantly increase the likelihood of human answers containing personal experiences (β = 0.030, p < 0.01) and opinionated insights (β = 0.024, p < 0.05). Additionally, there is a marginally significant increase in the likelihood of human answers offering alternative solutions (β = 0.015, p = 0.059). These findings suggest that human answerers can leverage their experiential knowledge and unique cognitive abilities to adjust their responses to complement or diverge from AI-generated answers.

Table 12. Human answerers' response to AI-generated answers

	(1)	(2)	(3)	(4)	(5)
	DV: text similarity between the first answer and the second answer		DV: whether the answer contains personal experience	DV: whether the answer contains opinionated insights	DV: whether the answer contains alternative solutions
AI_i	-0.034***	-0.086***	0.030**	0.024*	0.015
	(0.008)	(0.013)	(0.011)	(0.010)	(0.008)
question timing FE	√	√	√	√	√
controlling for LLM-enabled content quality ratings and length of the first answer		√			
controls	√	√	√	√	√
N	1,058	1,055	4,634	4,634	4,634

Notes: 1. ***p<0.001; **p<0.01; *p<0.05.
2. Robust standard errors are clustered at the question tag level.
3. Question timing FE indicates the date on which the question was posted.

7.3. Broader Platform Engagement: Effects of AI-Generated Answers on Questioner Activity and Question Visibility

Beyond our primary findings on human answer contributions and recognition, we examine whether AI-generated answers affect other important metrics of platform vitality: questioner retention, interactive dialogue between questioners and answerers, and overall question visibility.

These dimensions provide a more comprehensive understanding of AI's impact on the platform's knowledge exchange ecosystem.

Table 13 presents the effects of AI-generated answers on the questioner and viewer engagement metrics over a one-month period. Notably, we find no statistically significant effects across any of these engagement dimensions. Column 1 shows that questioners who receive AI-generated answers exhibit no significant change in subsequent posting activities ($\beta = 0.007$, $p > 0.05$), indicating that AI content does not measurably influence questioner retention.

In Column 2, we analyze questioner-answerer interaction by measuring comments posted by questioners in response to answers (limited to questions receiving at least one human answer). The results show no significant effect ($\beta = -0.043$, $p > 0.05$), suggesting that AI-generated answers do not change the depth of dialogue between questioners and answerers. Similarly, Column 3 reveals no significant difference in question viewership ($\beta = -0.009$, $p > 0.05$) between questions with and without AI-generated answers.

Taken together, these results suggest that the impact of AI-generated answers is largely confined to the dynamics of answer provision and recognition rather than affecting broader platform engagement. While AI integration creates significant changes in how human answerers contribute and receive recognition, it appears to have neutral effects on questioner retention, interactive dialogue, and question viewership.

Table 13. The impact of AI-generated answers on the engagement of questioners & viewers

	(1) DV: the logarithm of the number of questioner's subsequent activities after receiving an answer (*$Activity_i$*)	(2) DV: the logarithm of the number of comments the questioner posted in response to answers received by question *i* (*$Comment_i$*)	(3) DV: the logarithm of the cumulative viewership count for question *i* (*$View_i$*)
AI_i	0.007 (0.020)	-0.043 (0.025)	-0.009 (0.016)
question timing FE	√	√	√
other controls	√	√	√

N	3,613	2,748	3,613

Notes: 1. ***p<0.001; **p<0.01; *p<0.05.
2. Robust standard errors are clustered at the question tag level.
3. Question timing FE indicates the date on which the question was posted.

8. CONCLUSION

8.1. Main Findings

Our study investigates the impacts of platform-specific internal GenAI on user behavior in online Q&A communities. We show that AI-generated answers significantly reduce both the number of human answers and their recognition as measured by net votes. However, receiving AI-generated answers does not significantly impact questioners' subsequent activities, including posting new questions or providing comments. Question viewership also remains unaffected by the presence of AI-generated answers.

We identify three mechanisms underlying the negative impact on answer supply: the content effect, stemming from the quality of AI-generated content; the source effect, arising from the mere fact that the information source is an AI system; and the position effect, resulting from AI-generated answers appearing almost instantaneously, occupying the first position in answer sequences and consequently demotivating human contribution. Notably, our comparison of AI with high-reputation human answerers reveals that the source effect is unique to AI-generated answers—even the most reputable human answerers do not produce a similar deterrent effect.

The observed decrease in recognition of human answers cannot be attributed to a reduction in expert participation or declining answer quality. Rather, our analysis suggests that internal GenAI elevates viewers' quality expectations, creating a more demanding standard against which human contributions are evaluated.

8.2. Theoretical Contributions

Our study contributes to the emerging literature on generative AI's impact on online communities in several important ways. First, we extend research beyond external general-purpose GenAI tools to platform-specific internal GenAI systems. Our study offers unique insights into the challenges and opportunities platforms face when developing their own AI capabilities. We document a complex relationship between GenAI and human participation, which is especially relevant as more online platforms develop proprietary AI solutions in response to the proliferation of external GenAI tools.

Second, our study advances the theoretical understanding of the mechanisms through which GenAI influences user behaviors in online communities. We demonstrate the importance of examining GenAI's impacts through multiple theoretical lenses. Our study identifies multiple underlying channels through the lens of dual-system theories (Chaiken 1980) and expectation disconfirmation theory (Oliver 1980). The integration of these diverse theoretical perspectives enables a deeper understanding and more accurate interpretation of internal GenAI's effects. For example, while our initial results indicate negative effects on answer supply and recognition, our mechanism analysis reveals that there is no reduction in the proportion of expert answers or a decline in the underlying quality of human contributions. This insight is critically important for platforms seeking to develop a more comprehensive understanding of internal GenAI's effects.

Third, our findings about the source effect of AI-generated answers align with research that suggests AI can signal authority (Logg et al. 2019; Pakarinen & Huising 2023). This authority signaling occurs for multiple reasons. AI systems may appear to possess vast databases and computational power, which creates a perception of comprehensive and reliable answers. We show that in online technical Q&A communities, this source effect exists only for AI and not for human answerers—even highly reputable ones who produce comparable quality answers. Users appear to attribute unwarranted authority to AI systems. This insight has significant implications for human-AI interactions in online Q&A platforms and raises concerns about the potential over-recognition of AI in specific contexts.

Fourth, our analysis reveals that human answerers respond to AI-generated content by differentiating their contributions through unique, experience-based insights that AI cannot replicate. This finding contributes to ongoing discussions about AI's role in human-AI interactions (Bayer & Renou, 2024; Law & Shen, 2024; Wang et al., 2024). Unlike Chen and Chan's (2023) findings of an anchoring effect in creative tasks, our study suggests that AI integration might stimulate human creativity and diverse thinking in certain contexts. This observation creates opportunities for future research on optimizing complementarity between AI and human contributions in online communities.

8.3. Practical Implications

Our findings provide actionable insights for platform designers and community participants who navigate GenAI integration in technical Q&A communities. For platforms with a conservative approach, it may be prudent to delay GenAI deployment due to the observed negative effects on answer supply and recognition. This cautious approach allows platforms to maintain community dynamics while monitoring GenAI's evolving impacts.

Platforms that commit to GenAI implementation may consider targeted mitigation strategies. These include providing warnings about AI-generated answers' limitations, which can promote realistic user expectations. Additionally, platforms could introduce temporal delays in AI-generated answers, which would leverage position effects to encourage human contributions before AI responses appear.

For community members, our research highlights the importance of strategic differentiation when they contribute alongside AI. While GenAI excels at rapid, factual responses, human participants should emphasize their distinctive strengths in contextual understanding, experiential knowledge, and creative problem-solving. They can focus on areas where human cognition remains superior—such as real-world applications, nuanced interpretations, and innovative alternatives—to maintain their relevance and visibility.

This complementary approach between AI capabilities and human expertise preserves individual contributor value and enhances the collective knowledge ecosystem. Community members who take this differentiation approach can help sustain the vibrant exchange of specialized knowledge that defines successful technical Q&A platforms.

8.4. Limitations

While our work offers useful insights into the impact of GenAI on online technical Q&A communities, we acknowledge several limitations that present opportunities for future research. First, our study focuses on a technical Q&A community, where questions often have specific answers. The generalizability of our findings to non-technical settings, such as Quora, remains an open question. While existing literature has explored differences between social-oriented and technique-oriented platforms (Burtch et al., 2023), the specific impact of internal GenAI in these diverse contexts warrants further investigation.

Second, our experimental design compares scenarios with and without internal GenAI-generated answers. Future research could explore the effects of varying levels of control over GenAI's style and features. This would allow for a deeper understanding of how different GenAI characteristics influence user behavior and community dynamics.

Third, our analysis primarily focuses on the effects within a 150-day window. Research exploring the dynamic nature of these effects over an extended period would be valuable, especially considering the likelihood of longer-term coexistence between GenAI and human participants in online communities.

These limitations, rather than detracting from our findings, highlight promising avenues for future research. By addressing these areas, subsequent studies can build upon our work to provide a more comprehensive understanding of GenAI's role in diverse online communities, its long-term impacts, and strategies for effective human-AI collaboration.

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APPENDICES

Appendix A: Background Information on the Randomized Experiment

The platform's randomized experiment appears directly linked to the release of ChatGPT. Nine days after ChatGPT's public release on December 9, 2022, SegmentFault implemented a policy prohibiting ChatGPT-generated answers on its platform.[8] This decision likely responded to community concerns about reliability, accuracy, and the absence of human verification in AI-generated content. The platform enforced this policy through user reporting and automated detection systems while permitting answers that incorporated human verification and proper citation of AI sources.

Nine months after implementing this policy, the randomized experiment was initiated, likely driven by the intent to explore potential avenues for leveraging the increasingly prevalent and sophisticated GenAI tools. It is plausible that the platform recognized the need to strike a balance between harnessing the capabilities of these technologies and addressing the concerns raised by its community regarding the reliability and credibility of AI-generated content. Through this experimental approach, SegmentFault aims to gather empirical data and user feedback, which will inform its decision-making process and guide the development of strategies for integrating generative AI in a manner that enhances the platform's value proposition while maintaining the integrity and trustworthiness of the community-generated content.

Table A. Random assignment test results

		DV: treatment status			
		Bivariate Regressions	Multivariate Regression		
		(1)	(2)	(3)	(4)
Questioner	log reputation score	-0.006	-0.008	-0.009	-0.010
		(0.004)	(0.006)	(0.006)	(0.006)

[8] Although ChatGPT is officially unavailable in China, people in China may access ChatGPT via VPNs.

	number of gold medals earned	-0.001	0.001	0.001	0.001
		(0.001)	(0.003)	(0.003)	(0.003)
	number of silver medals earned	-0.000	-0.001	-0.001	-0.001
		(0.000)	(0.001)	(0.001)	(0.001)
	number of bronze medals earned	-0.000	0.000	0.000	0.000
		(0.000)	(0.000)	(0.000)	(0.000)
Question Content	log number of Chinese characters	0.014		0.012	0.013
		(0.008)		(0.008)	(0.008)
	presence of photos	-0.006		-0.005	-0.008
		(0.017)		(0.017)	(0.018)
	presence of hyperlinks	0.026		0.024	0.020
		(0.029)		(0.030)	(0.030)
	presence of tables	0.027		0.011	0.012
		(0.063)		(0.066)	(0.066)
	presence of block quotes	0.095		0.118	0.116
		(0.145)		(0.148)	(0.149)
Question Type	web and mobile development	0.003			-0.009
		(0.017)			(0.026)
	software design and architecture	-0.042			-0.065
		(0.082)			(0.087)
	cloud computing	-0.045			-0.069
		(0.039)			(0.046)
	databases	0.006			0.009
		(0.053)			(0.058)
	computer networks and systems	0.107			0.061
		(0.083)			(0.088)
	AI and machine learning	0.062			0.067
		(0.079)			(0.084)
	development tools and environment	0.039			0.034
		(0.048)			(0.053)
	programming languages	-0.020			-0.031
		(0.023)			(0.033)
	career and professional development	-0.045			-0.026
		(0.118)			(0.124)
	question timing FE (date on which the question was posted)		√	√	√
	N		3,613	3,613	3,613
	Adjusted R-squared		0.003	0.003	0.002

Notes: ***p<0.001; **p<0.01; *p<0.05.

Appendix B: LLM-Enabled Content Quality Rating

In designing prompts to assess the content quality of answers in an online Q&A community, we employed a comprehensive approach informed by best practices in prompt engineering, particularly drawing from the OpenAI handbook.[9] Our methodology incorporated several key strategies to enhance the reliability and effectiveness of the assessment process.

Central to our approach was an emphasis on clear instructions. We provided detailed queries to elicit relevant and focused responses from the language model. Specifically, we used Kimi, a prominent LLM developed by Moonshot, a leading vendor in China, to evaluate the overall content quality. To mitigate potential misinterpretations, we conducted thorough validations of the prompts. We implemented a well-crafted system message to establish a specific persona for the model, ensuring consistency in its role throughout the assessment. To further structure the input and output, we employed clear delimiters and specified the desired output format. Additionally, we incorporated a chain-of-thought approach to delineate the steps required to complete the assessment task. This not only made the evaluation process more transparent but also allowed for a more nuanced analysis of the model's reasoning. To maintain concision and focus, we requested refined reasons for outputs to be provided in single sentences.

A crucial aspect of our prompt design was giving the model sufficient time to "think." We instructed the language model to generate its own solutions before evaluating the provided answers. Furthermore, we employed techniques such as inner monologue or sequential queries to ensure that the model's reasoning process remained hidden, preventing the output of unnecessary content. Our assessment framework comprised two main components: the system prompt and the user prompt. The system prompt defined the role and behavior of the language model. The user prompt contained specific task instructions and the content for evaluation, directing the model's engagement with the material at hand. The prompts used for the assessment task are detailed below:

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[9] https://platform.openai.com/docs/guides/prompt-engineering/prompt-engineering.

System:

Use the following step-by-step instructions to evaluate the technical question, measuring how well it solves a problem objectively without any emotion.

Step 1: The user will ask for evaluating the objective answer quality. First work out your own answer to the problem. Don't rely on the provided answer as it may not be correct.**Do not output all your work for this step**.

Step 2: Compare your answer to the provided answer. Objectively evaluate the quality of the provided answer by considering the following dimensions: clarity, readability, accuracy, relevance, level of detail, personal experiences, personal insights, innovative approaches, alternative solutions, and engaging storytelling.**Do not output all your work for this step**.

Dimensions Definitions:
1.**Clarity**: The ability to convey information in a way that is easily understandable, avoiding ambiguity and confusion.
2.**Readability**: The quality of being easy and pleasant to read, based on factors such as sentence structure, word choice, and answer formatting.
3.**Accuracy**: The state of being precise, correct, and in conformity with fact or truth, without errors or inaccuracies.
4.**Relevance**: The degree to which the provided information is applicable, connected, and pertinent to the specific topic or question at hand.
5.**Level of detail**: The amount and specificity of information provided, ranging from a high-level overview to an in-depth explanation with supporting facts and examples.
6.**Personal experiences**: The incorporation of personal anecdotes or real-world examples that are specific to the individual answerer and not commonly known or shared.
7.**Personal insights**: The ability to provide personal opinions, preferences, and recommendations.
8.**Innovative approaches**: The suggestion of creative, original, or unconventional ideas, solutions, or methodologies that deviate from traditional or widely accepted norms in problem-solving.
9.**Alternative solutions**: The ability to suggest alternative approaches or solutions that may not be directly related to the original question but can still help solve the underlying problem.
10.**Engaging storytelling**: The ability to present information or ideas in a compelling, narrative form that captures the reader's attention, makes the content more memorable, and enhances understanding or relatability.

Step 3: Reply with an overall quality rating and quality rating for each dimension**in required format shown below**. Each score should be an integer from 1 to 10, with 1 denoting extremely low quality and 10 denoting extremely high quality.**After finishing all scorings**, explain the basis for scores of the 10 dimensions within**one sentence**.

Required Output Format:
Overall quality: 1 to 10
Clarity: 1 to 10
Readability: 1 to 10
Accuracy: 1 to 10
Relevance: 1 to 10

Level of detail: 1 to 10
Personal experiences: 1 to 10
Personal insights: 1 to 10
Innovative approaches: 1 to 10
Alternative solutions: 1 to 10
Engaging storytelling: 1 to 10
Basis for scores: briefly explain the basis for the overall quality score and 10 scores of dimensions here in one sentence

User:

You need to objectively evaluate the quality of the given answer based on the following dimensions: clarity, readability, accuracy, relevance, level of detail, personal experiences, personal insights, innovative approaches, alternative solutions, and engaging storytelling. Provide a quality score from 0-10 for overall quality and then each dimension**in required format shown below**, with 0 being extremely low quality and 10 being extremely high quality.**After finishing all scorings**, explain the basis for scores of the 10 dimensions within**one sentence**.

Here is the**required Output Format**:

Overall quality: 1 to 10
Clarity: 1 to 10
Readability: 1 to 10
Accuracy: 1 to 10
Relevance: 1 to 10
Level of detail: 1 to 10
Personal experiences: 1 to 10
Personal insights: 1 to 10
Innovative approaches: 1 to 10
Alternative solutions: 1 to 10
Engaging storytelling: 1 to 10
Basis for scores: briefly explain the basis for scores of the 10 dimensions within one sentence

We conducted several tests to validate the LLM-enabled content quality ratings. First, we examined cases where the internal GenAI was unable to provide a substantive answer, typically responding with a statement like “I’m sorry, but I can’t give a detailed answer because of [reason].” If the LLM-enabled content quality ratings were valid, these failed answers should receive lower ratings compared to other answers. A t-test confirmed that failed AI answers indeed received significantly lower ratings than other answers ($t = -3.749$, $p = 0.001$).

Second, we compared the LLM-enabled content quality ratings between human answers accepted by questioners and those not accepted. We expected accepted human answers to receive higher ratings. This expectation was supported by a t-test, which showed that LLM-enabled content quality ratings for accepted human answers were significantly higher than those for non-accepted answers ($t = 8.784$, $p = 0.001$).

Third, we investigated the relationship between LLM-enabled content quality ratings and recognition by viewers. Specifically, we used net likes to quantify recognition by viewers and divided answers into two groups based on the median net likes. Comparing these two groups, we found that answers with above-median net likes consistently received higher LLM-enabled content quality ratings. This difference was statistically significant ($t = 7.532$, $p = 0.001$). This result further supports the validity of the LLM-enabled content quality ratings.